TurnoutBoss User Guide

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Abstract

This is the user guide for TurnoutBoss boards. It discusses basic installation and configuration of the boards to control turnouts and signals around a single, simple passing siding.

1 Introduction

The TurnoutBoss is designed to provide a simple implementation for controlling and signaling simple passing sidings. The emphasis here is on simple, so that a new LCC user can easily configure, install and operate the board.

The TurnoutBoss is a LCC/OpenLCB board that contains:

- A driver for a stall-motor turnout motor, e.g. a Tortoise or similar,
- A driver for a servo turnout motor,
- Three CT-based DCC current occupancy detectors with adjustable sensitivity and status indicator LEDs,
- An input for a optical sensor to supplement the CT-based sensor on the turnout,
- Two inputs for turnout-control switches which can be used in several modes,
- An input for a digital sensor/indicator "touch button" to control the turnout,
- Two inputs for optional turnout position feedback using one or two switches,
- Four signal-head drivers for common anode, common cathode or bidirectional LED three-color signal heads with adjustable intensity, with each head having connections for red/yellow/green lamps and a lunar lamp,
- An LCC interface for communications, control and configuration, with an optional on-board terminator,

- Buttons and LEDs for local configuration,
- And signaling logic to run those signals.

The board's use of LCC allows simple connections between boards to exchange the occupancy and signal-state information needed to do prototypical signaling. It also allows the TurnoutBoss board to work with separate display and control panels without any additional wiring.

1.1 Signaling Around A Simple Siding

Figure 1: Example of a railroad with signals around a simple passing siding. Signaling the center siding would use two TurnoutBoss boards.

In its simplest form, signals tell the engineer one of three things:

1. "Clear to run at full speed in the next block" - this is the "Clear" indication, usually denoted by a green appearance.
2. "Approach the next signal ready to stop" - this is the "Approach" indication, usually denoted by a yellow appearance
3. "Stop before this signal and don't proceed" - this is the "Stop" indication, usually denoted by a red appearance

The TurnoutBoss implements this simple, but realistic, version of signaling. The Stop indication is used when the next block is occupied, or a turnout is set against the forward path. The Approach indication is used to warn the engineer that the next signal is at Stop, so the train can be slowed down and prepared to brake to a stop.

When entering a turnout from the facing points, most signal systems give an indication to the engineer whether the train will be taking the Normal/mainline or Diverging/siding path. The speed limits may be different through the two routes, for example. This is commonly done with a double-headed signal where the top head represents the indication for proceeding on the Normal path, and the lower head for proceeding on the Diverging path.[1]

The TurnoutBoss by default expects a double-headed signal to be wired for the facing point position. It can also be configured to properly drive a single-headed signal there if you'd prefer that.

2 Planning Your Installation

In this section, we discuss planning your installation to save time later when configuring and wiring your boards.

To simplify configuration, the TurnoutBoss boards need to be wired to your layout a specific way. One board is associated with each turnout and the signals around it.

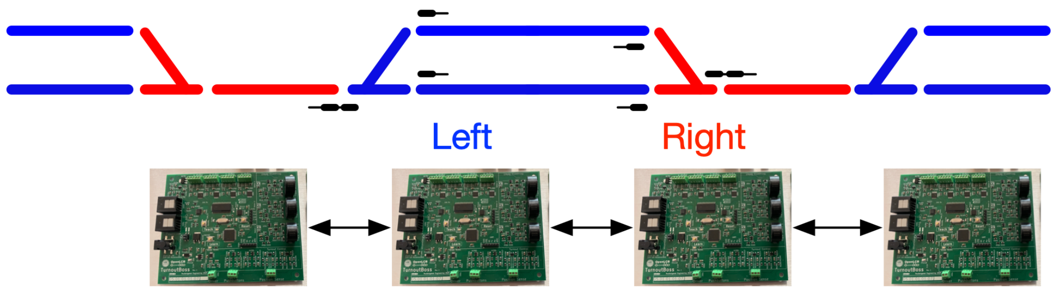


Figure 2: TurnoutBoss boards controlling the signals around four turnouts on our sample railroad. The boards at the two ends of the turnout are configured as "Left" and "Right" mode, respectively, so they understand the type of track to which they're connected. The arrows indicate that they're communicating with each other via the layout's LCC network.

You start by assigning a board to each turnout that you want to signal.

In order to handle detection of track occupancy (see below), the boards are wired differently depending on whether they're controlling the "Left" or "Right" end of a siding.

[1]Formally, the appearance of the two heads is given a single name. Red over yellow, for example, will be called "Diverging Approach"; Red over green will be called "Diverging Clear".

2.1 How do you want to control your turnouts?

If you want to provide local control of your turnout from the fascia, there are two approaches:

1. You can have one pushbutton which alterates the turnout position between Normal and Diverging. This is the default behavior.
2. You can have two pushbuttons: One sets the turnout to Normal (straight-ahead) and the other sets the turnout to Diverging.

Once you've picked one of these, you'll configure the TurnoutBoss to use it and then make the necessary wiring connections.

You can also control the turnout remotely from another board or computer via the LCC network. See the Appendix for the event IDs you'll use to configure that into the remote board or computer.

If you only use one input to toggle the switch position, you can use the other connection as a general purpose input. It'll generate LCC events that you can configure to operate other boards. See the Appendix for more on that.

2.2 Setting up your occupancy blocks

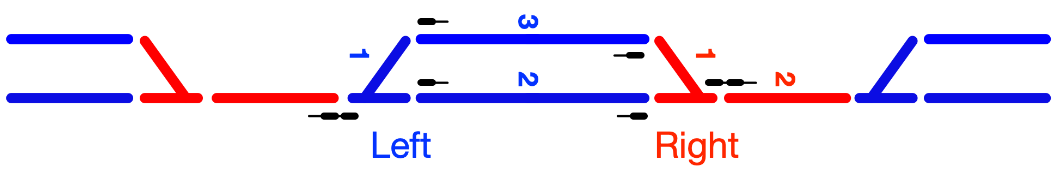


Figure 3: Wiring of occupancy detectors on multiple boards. The tracks marked in blue are connected to the left-hand board. The left turnout is connected to detection connector 1, the main between turnouts is wired to detection connector 2, and the siding is wired to detection connector 3 on the left-hand board. The boards marked in red are connected to the right-hand board. The right turnout where the board is located is wired to detection connector 1 and the main between sidings is wired to detection connector 2 on the right-hand board.

As we've seen above, proper operation of signals depends on knowing which section(s) of track are occupied by rolling stock.

The TurnoutBoss uses current-sensing occupancy detectors to do this. You isolate one rail in each section of track, then feed power to that section through the occupancy detection coil on the TurnoutBoss. When a DCC decoder-equipped locomotive or rolling stock with resistor wheelsets draws even a little current through the rails, the detector goes active to tell the TurnoutBoss logic that the related track block is occupied.

Signals will be placed even with the gaps between the blocks so they can protect entry to the next block. This means that the gaps should be cut at the place where you want trains to stop. Typically, this is at the clearance points on the frog side of the turnout, and just before the turnout's points on the entry side.

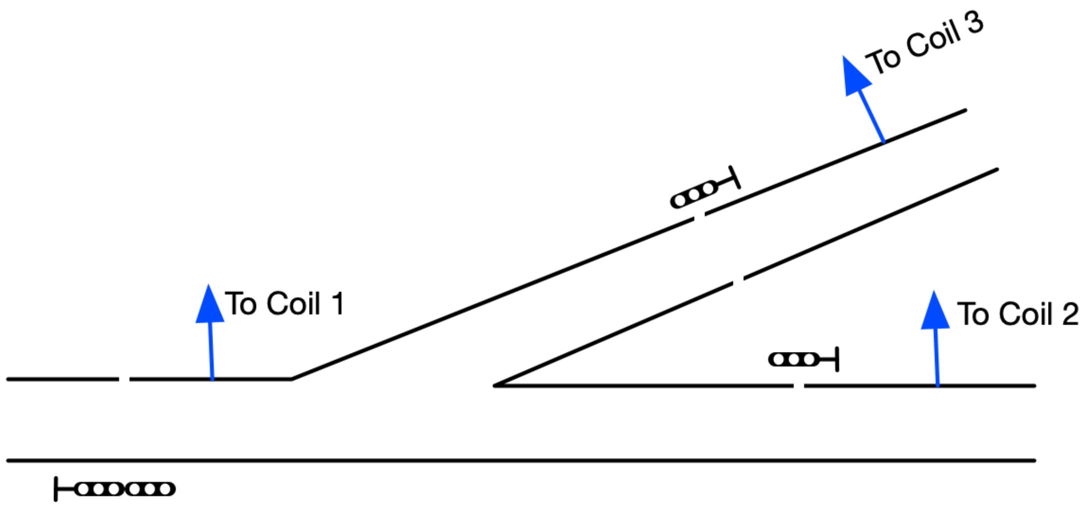
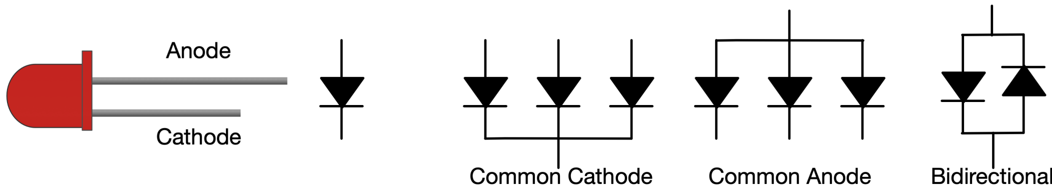


Figure 4: Gap and signal locations around a left-end turnout. Each block between the gaps has one rail fed through the proper occupancy detector coil on the the TurnoutBoss board. An optical sensor can also be installed in the turnout to detect rolling stock that doesn't draw current.

2.3 Wiring your signals

Signals commonly come wired in one of three ways. The TurnoutBoss can be configured to work with any of these.



1. Common cathode - the three cathodes (bar side of the diode symbol) of the three Red/Yellow/Green LEDs are connected together and brought out as a single wire.
2. Common anode - the three anodes (triangle side of the diode symbol) of the three Red/Yellow/Green LEDs are connected together and brought out as a single wire.
3. Bidirectional LED - there are only two leads brought out. When one lead is positive, the LED shows Red. When the other is brought out, the LED shows Green. To show Yellow, the driver alternates rapidly between Red and Green.

It's best if you have all the signals wired the same way using common anode or common cathode, because then the basic button-based configuration can select any of those for all attached signals at the same time.

The TurnoutBoss controls the current, so you shouldn't have resistors in series with your signal LEDs.

If you do need to use bidirectional LEDs or multiple types of signal wiring, you can use a computer-based configuration tool to set that up within the TurnoutBoss board. See below.

Generally, prototype signaling systems have a double-head signal controlling access to the points of the turnout. That's the TurnoutBoss default. If for some reason you need to place a single-head signal there, you'll configure the TurnoutBoss to properly control that using a configuration tool. See below.

Figure 5: The four signal heads around a single turnout. You'll wire your signal heads to the driver circuits on the TurnoutBoss board as indicated here. Note that head "c" is the upper head and "d" is the lower head on a single mast. The "d" head is optional, although it's recommended for prototypical signaling.

2.4 Do you want turnout feedback?

Most model railroaders just rely on the turnout mechanisms to work properly: If they've commanded a turnout to move, the expect it will actually move to the commanded position. These people should skip this section.

Some model railroaders wire their railroad to have "feedback" on whether the points on their turnouts actually moved. The TurnoutBoss can do this, but it requires additional switches and wiring.

This extra wiring is not so bad if you just want to sense that the turnout motor has moved. For example, there are extra switch contacts on a Tortoise switch machine that can be connected to the TurnoutBoss board to sense that the machine moved.

If you want to check the actual position of the points, to make sure that e.g. a throw wire hasn't jammed or a bit of gravel hasn't blocked them, you can attach one or two microswitches to the points and connect them back to the TurnoutBoss.

You configure the TurnoutBoss to use zero, one or two feedback connections from the turnout.

- With one connection or switch, you can tell the position of the turnout's points.
- With two connections or switches, you can tell when the points are moving and that the points have made up properly on each side.

You'll configure the TurnoutBoss to this choice below. The default is to not use the feedback inputs.

If you don't use one or both of the inputs for turnout feedback, you can instead use them as general purpose LCC inputs. If you connect a switch to them you can have them emit general-purpose LCC events, which you can use to control operation of other boards. For more on this, see the Appendix.

2.5 How do you want to provide power to the boards?

A TurnoutBoss consumes about 110mA of current at 12V to 15V in normal operation, including the current needed for the signal lamps and slow-motion switch machine output. (See below for information on power for the servo output.)

There are three ways to provide the needed power to the TurnoutBoss boards:

1. The board can get it's power from the LCC network cable. If some other node or device is providing sufficient power to the LCC cable, the TurnoutBoss will take the power it needs from the cable.

2. The board can be connected to a 12V to 15V wall-wart power supply through the barrel connector on the board.

   When powered this way, the board can supply up to 500mA of current to each of its LCC connectors. Each of these can be connected to up to four other TurnoutBoss boards to power them.

3. The board can be connected to a 12V to 15V power supply through the power input terminals on the board.

   When powered this way, the board can supply up to 500mA of current to each of its LCC connectors. Each of these can be connected to up to four other TurnoutBoss boards to power them.

For example, you might have six TurnoutBoss boards operating three sidings. If you provide power from a wall-wart transformer to one of tne center boards, it can in turn provide power through the LCC cable to the other TurnoutBoss boards.

The servo output requires its own power supply. The servo your using will specify its input voltage. The TurnoutBoss requires that this be between five and eight volts.

3 Basic Configuration of the TurnoutBoss Boards

We discuss basic configuration before the Wiring section because it's easier to configure your TurnoutBoss boards on your workbench than while lying on your back under the layout. We'll set them up using the decisions you made in the "Planning Your Installation" section above.

3.1 Preparation

Label the boards with a piece of tape so you know which board goes where on the layout.

Then connect them into a small LCC network and power them up:

1. Connect your TurnoutBoss boards with CAT-5 (Ethernet) cables to create a local LCC network.
2. Put a CAN Terminator in each of the empty connectors on the two boards at the end of that LCC network.
3. Plug a 12V to 15V power supply (wall-wart) into one of the TurnoutBoss boards and plug that power supply into wall power.

The green LED on each of the boards should come on. If not, check your power supply and that the LCC cable is properly connecting the boards.

If you press and release the Reset button on one board, the yellow LED(s) on the other board(s) should blink along with the blue and yellow LEDs on the board you reset. If not, check that you have the proper cable type and have the terminators in place.

3.2 Reset the TurnoutBoss boards to factory defaults

Resetting the boards to factory defaults ensures that you're starting configuration from a well-defined state.

To do that, on each board:

1. Hold down the Teach and Learn buttons
2. Press and release the Reset button
3. When the green LED comes on, release the Teach and Learn buttons
4. Press the Teach button one more time

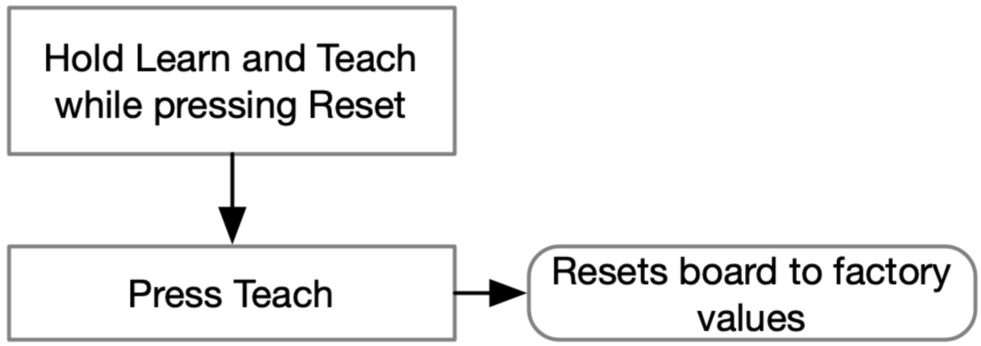


Figure 6: Organization of the button presses to reset the board to factory defaults.

3.3 Configure the Connections between Boards

In this step, you'll use button presses on the board to configure the connections between your TurnoutBoss boards.

If you're planning on attaching a configuration tool, see below, you can skip this step and do it later with the configuration tool. That may be easier.

It's also possible to do the button pushes for this configuration again once the board is installed on the layout, although it might be less convenient to access the boards then.

In this step, you'll configure:

- Which board is to the left and right of this board
- Whether the board is at the left or right end of the siding.

See the figure below for the sequence of button pushes needed to configure these. There are no defaults for these settings. You have to go through and set the the left and right neighboring boards, and whether this board is implementing a left or right end of the siding.

You start by holding the Learn button, pressing Reset, then promptly releasing the Learn button. While in this mode, the board will be flashing its green LED.

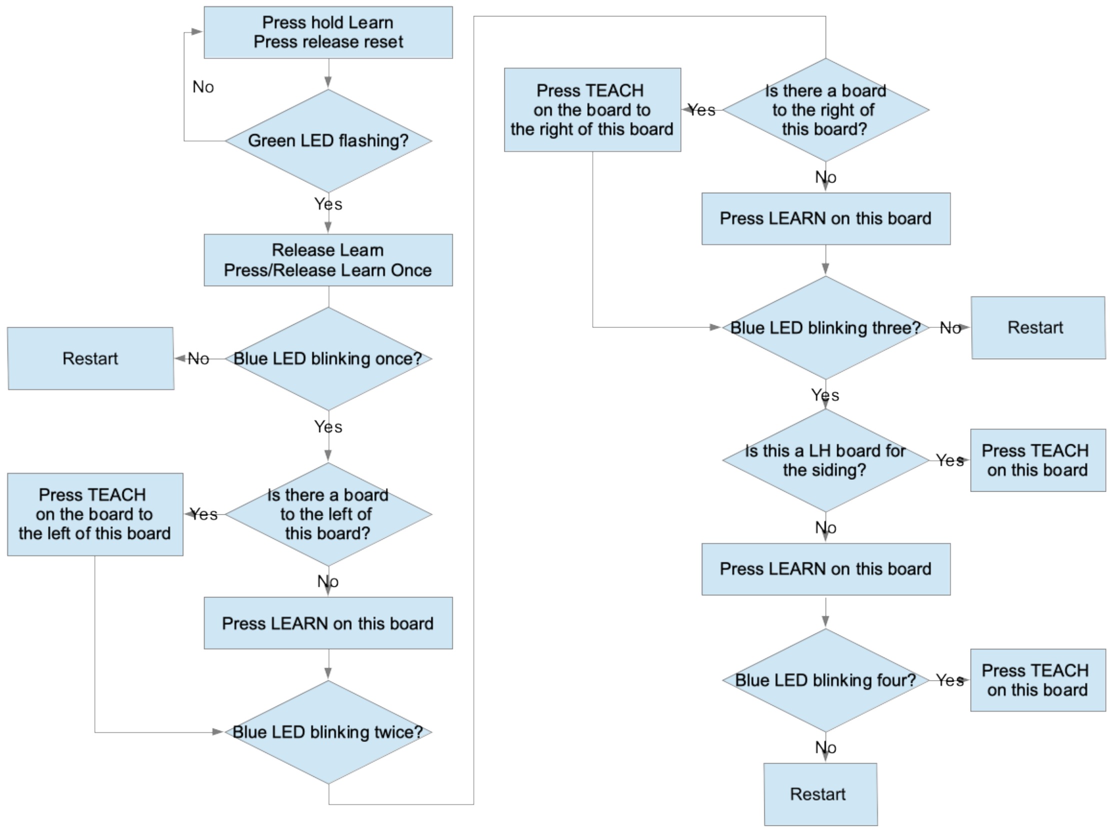


Figure 7: Organization of the button presses to configure board connections.

3.4 Configure the Initial Hardware Options

In this step, you'll use button presses on the board to configure some initial hardware options, if you need something other than the default values.

If you're planning on attaching a configuration tool, see below, you can skip this step and do it later with the configuration tool. That may be easier.

It's also possible to do the button pushes for this configuration again once the board is installed on the layout, although it might be less convenient to access the boards then.

In this step, you'll configure:

- How the signal heads are wired,
- Whether the turnout position is controlled by one or two attached buttons.

See the figure below for the sequence of button pushes needed to configure these.

The defaults are common-cathode wiring and one-button control of the turnout. If those are your choices, and if you have done the reset above, you don't have to configure those settings.

You start by holding the Learn button, pressing Reset, holding the Learn button until the yellow light comes on, then releasing the Learn button. While in this mode, the board will be flashing its green LED and the yellow light will remain on.

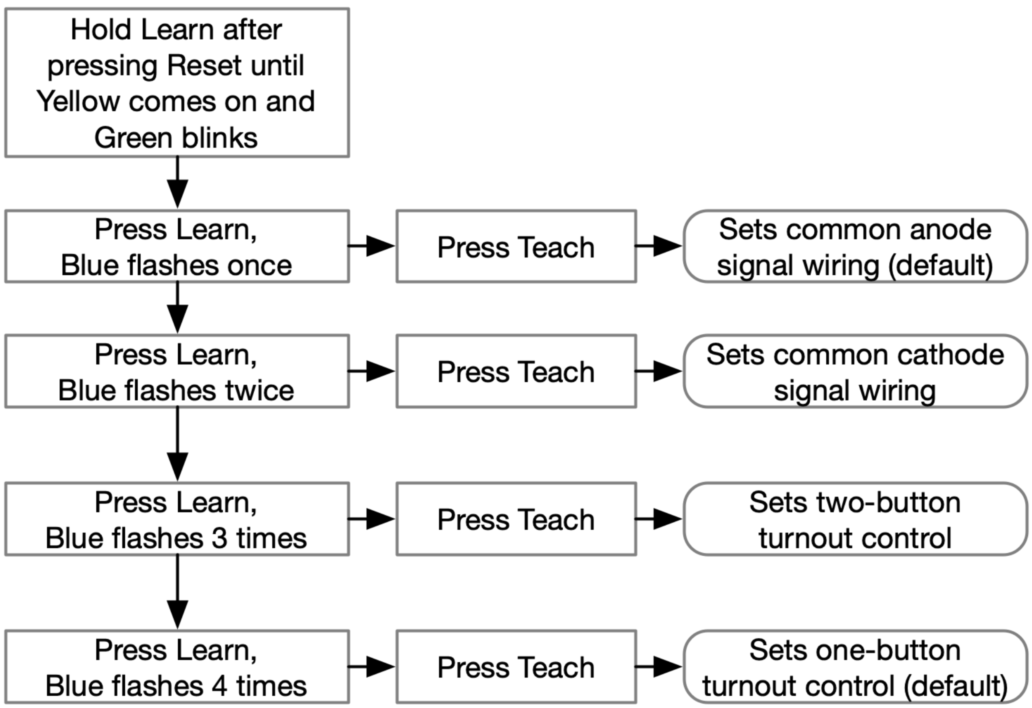


Figure 8: Organization of the button presses to configure initial hardware options.

4 Installation and Wiring

First, attach your TurnoutBoss boards to your layout in a convenient location using spacers, screws and the mounting holes in the corners of the boards.

4.1 LCC Cable and Power

If you have an existing LCC network, connect the boards into it using CAT-5 (Ethernet) cables. If these are the first LCC boards on your layout, make sure there are terminators in the open RJ-45 jacks on the end boards.

The TurnoutBoss boards need to have power supplied. If there's sufficient power available on the LCC network from some other source, you don't need to do anything more; the boards will take the necessary power from the LCC cable. If not, or if you're not sure, attach a 12V to 15V power supply (wall wart) to the power connector on one of your TurnoutBoss boards. That can power other TurnoutBoss boards through the LCC network.

4.2 Turnout Motor and Pushbuttons

Wire the + and - output on the Driver connector to the motor contacts on your Tortoise or similar low-current turnout motor.

To provide local control of your turnout from the fascia, there are two approaches:

1. You can have two pushbuttons: One sets the turnout to Normal (straight-ahead) and the other sets the turnout to Diverging. To wire this, connect a SPST normally open pushbutton between the N and Gnd spots on the Pushbutton connector, and connect another pushbutton between the D and Gnd spots on that same connector. You should have configured this option in the configuration section above, or you can do that now.

2. You can have one pushbutton which alterates the turnout position between Normal and Diverging. To wire this, connect a SPST normally open pushbutton between the N and Gnd spots on the Pushbutton connector. You should have configured this option in the configuration section above, or you can do that now.

If you haven't configured the turnout control buttons, the board defaults to one-button configuration, so regardless of whether you have one or two buttons attached, pressing the Normal button should flip the turnout position.

If you have a touch-button input, wire it to the TurnoutBoss connector now.

4.3 Servo Output

The servo output requires a separate power supply for reliable operation. Your servo will determine what voltage is needed, but it has to be between 5 and 8 volts.

Connect the three-wire servo cable to the connector on the board with the proper polarity. If it's connected backwards, the servo won't operate, but nothing will be harmed.

The default starting position for the servo output when set to normal is 85 of 255. The default starting position for the servo output when set to diverging is 170 of 255.

You can adjust the starting and final position of the servo using a configuration tool. See below.

4.4 Occupancy Detectors and Optical Input

Run the power feed wire for each of the occupancy sections through the hole in the corresponding coil. See the labeling on the board for coil goes with which block feeder. It doesn't matter which way the wire goes through the hole in the coil. If you determine that you need additional sensitivity, you can pass the wire through the coil twice, so that it makes a complete turn wrapped around the coil

To test this, put a locomotive or resistor-equipped car on each section of track; the corresponding LED on the TurnoutBoss board should light and go out when the rolling stock is removed. Be sure to check adjacent sections don't light the LED to make sure you have the rail gaps in the right places.

You can adjust the sensitivity of the individual detectors using a configuration tool. See below.

If you have an optical sensor on your turnout, wire it to the TurnoutBoss connector. By default, an active input means that the turnout is occupied. You can invert this, so that an inactive input means that the turnout is occupied, using a configuration tool. See below.

4.5 Signals

The board defaults to common anode signal wiring. If you have common cathode, you'll have set the board configuration above. If you have bidirectionally-wired signals, you'll set that using a configuration tool. See below.

See Figure 5 above for which signal is wired to which connector.

- If you have common anode signals, connect their common connection to the A terminal on the appropriate signal connector, then connect the individual LED leads to the R/Y/G connections on that signal connector.

- If you have common cathode signals, connect their common connection to the K terminal on the appropriate signal connector, then connect the individual LED leads to the R/Y/G connections on that signal connector.
- If you have bidirectional signals, connect their leads to the R and G connections respectively on the appropriate signal connector.

The board has a signal-wiring test mode, entered by holding Teach and pushing Reset. It blinks the LEDs on each head in order: Red, then Yellow, then Green. Head A blinks its lamps, then Head B, then Head C, and finally Head D. Press Reset to return to normal operation.

You can adjust the intensity of the signal LEDs using a configuration tool. See below.

4.6 Turnout Feedback

If you decided to use turnout feedback, you have to wire it.

- If you selected one-switch feedback, connect a switch that's closed when the turnout is in the Normal position between the N and Gnd connections on the Turnout Feedback connector.
- If you selected two-switch feedback, connect a switch that's closed when the turnout is in the Normal position between the N and Gnd connections on the Turnout Feedback connector. Then connect a separate switch that's closed when the turnout is in the Diverging position between the D and Gnd connections on the Turnout Feedback connector.

5 Using a Configuration Tool

In addition to all the ones you can configure via the buttons and LEDs, there are more items that you can configure on the TurnoutBoss board You access these via a configuration tool. A configuration tool is a computer-resident tool that talks to a TurnoutBoss board and presents menus you can use to configure options.

Commonly used configuration tools include

1. JMRI - for PC, Linux and Mac computers [2]
2. Model Railroad System - for PC, Linux and Mac computers [3]
3. LccTools - for iPhone, iPad and Mac computers [4]

Please consult their web sites for instructions on how to install those tools, how to connect them to your LCC network and how to operate them. The figures below were made using JMRI, but the other tools provide similar configuration pages.

You can tell which board you're configuring by looking at the blue LED on the board. It will blink when you're communicating with that board.

The configuration information is in three segments:

- Layout Configuration Setup
- Hardware Configuration
- Producers and Consumers

We describe the contents of these below.

[2] Available from [https://jmri.org]
[3] Available from [https://www.deepsoft.com/home/products/modelrailroadsystem]
[4] Available from the Apple App Store or via [https://apps.apple.com/us/app/lcctools/id1640295587]

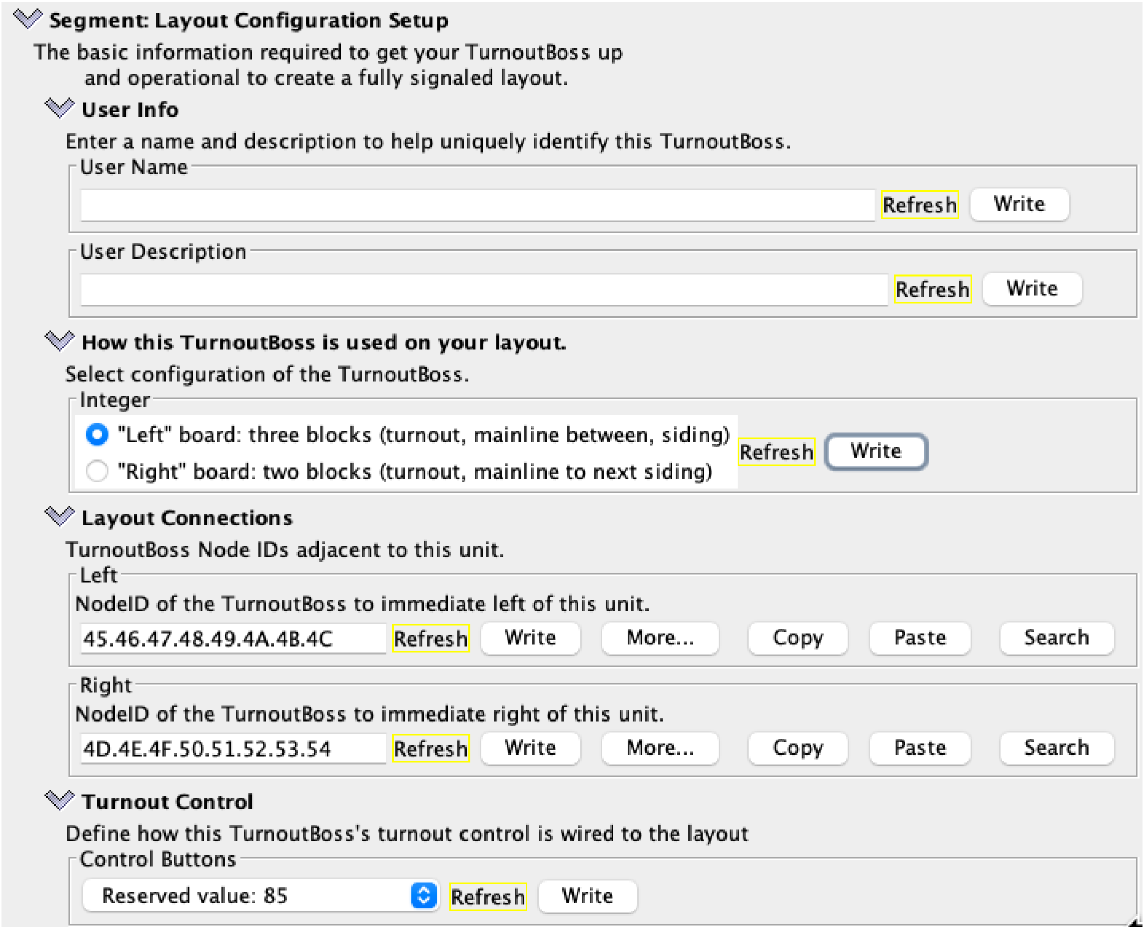


Figure 9: Layout Configuration Setup Section

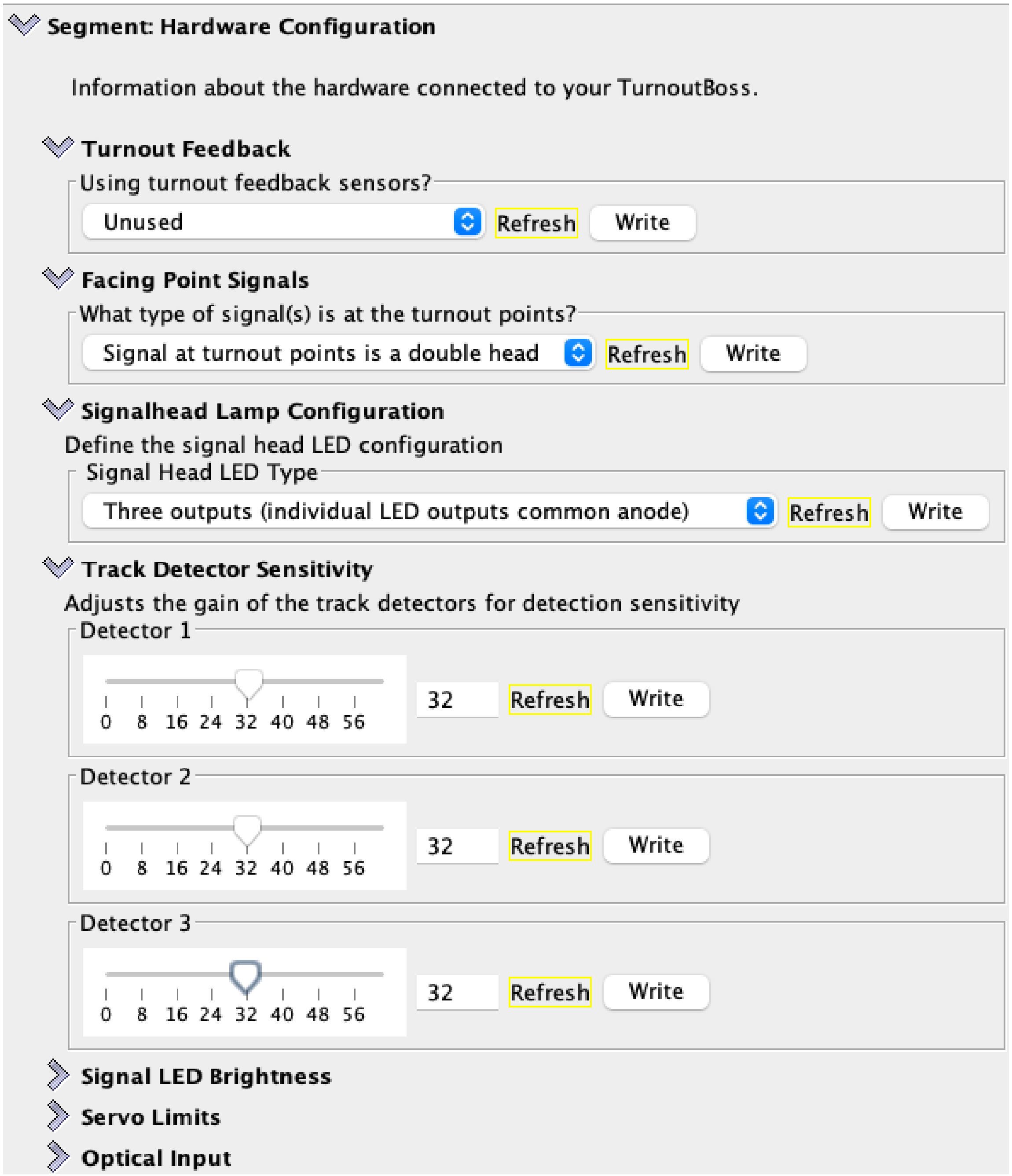


Figure 10: Hardware Configuration Section part 1

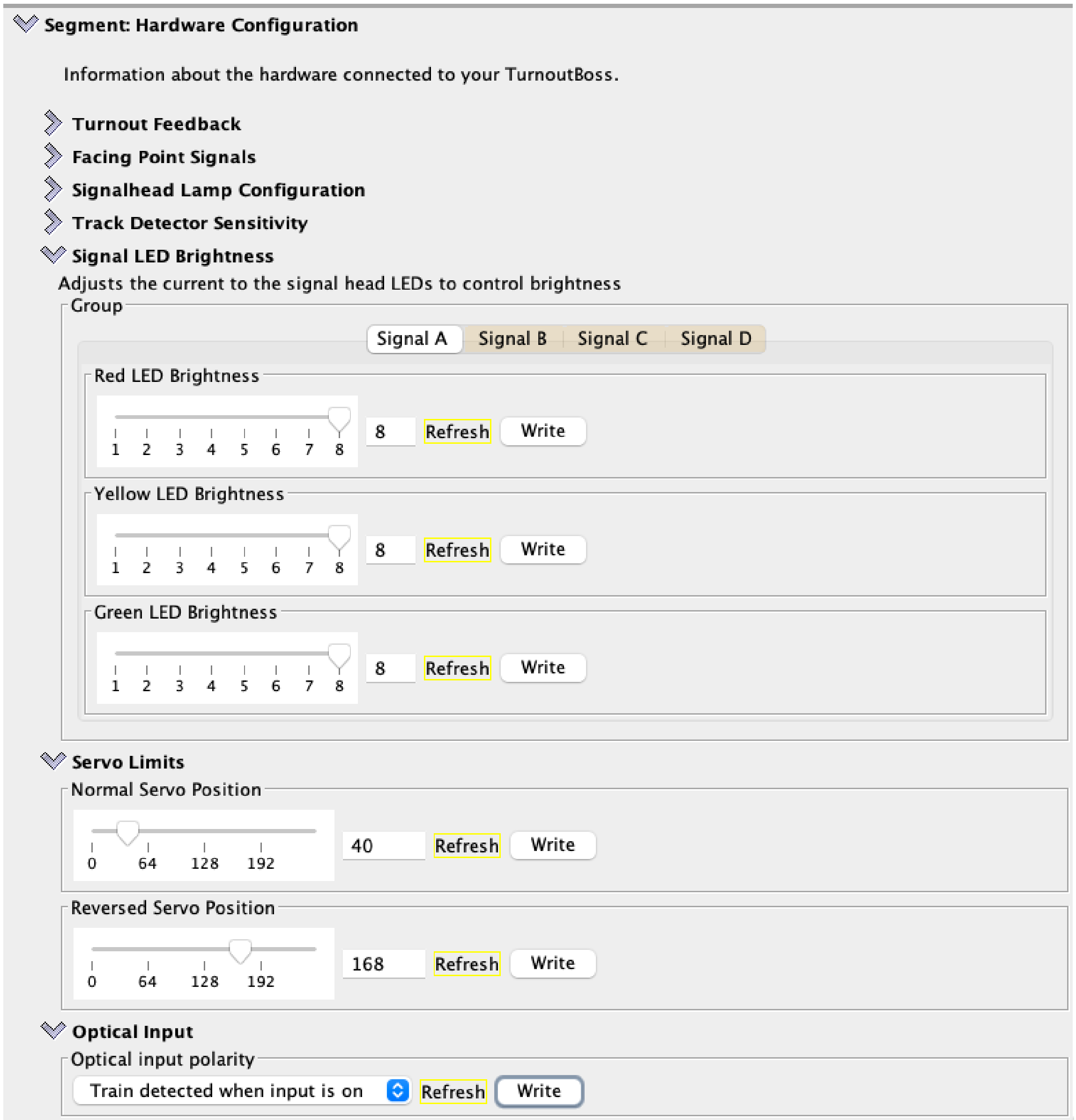


Figure 11: Hardware Configuration Section part 2

5.1 User Info

Once you have a configuration tool attached, the first thing you should do is to set the board's User Name and User Description. This will make it easier to identify the board and configure it later.

5.2 Board Usage

If you haven't configured the board configuration for use on the "left" or "right" end of the siding, you can do that here.

5.3 Layout Connections

You can enter the node ID of the boards to the immediate left and right. These are entered as an eight-byte string, with the first six being the node ID of the board, and the last two being zero.

5.4 Turnout Control

This selects whether the turnout position is controlled by zero, one or two attached buttons.

5.5 Turnout Feedback

This selects whether you're using turnout feedback, and if you are, whether you are using one input or two. The default is no turnout feedback.

5.6 Facing Point Signals

This selects whether the signal at the facing points has one head or two. The default is two heads.

5.7 Signalhead Lamp Configuration

You can select whether your three-LED signal heads are wired common anode or common cathode or whether you're using bidirectional-diode signal heads. Note that this applies to all the signal heads you have attached.

5.8 Track Detector Sensitivity

From the factory, the sensitivity of the TurnoutBoss board's occupancy sensors is set to about 1 mA. This will detect a single 10k resistor on a car. You might find that you need to adjust this sensitivity up or down due to the conditions on your layout. You can do that in this section. Each occupancy detector can be independently adjusted. The factory setting is 32 of 64. Higher numbers are less sensitive and lower numbers are more sensitive.

The LED associated with each channel on the TurnoutBoss board lights when occupancy is detected and goes off when no occupancy is detected. One way to set the sensitivity is to set the value high enough that the LED goes on, then lower it by a few steps until the LED goes out, then lower one or two more steps.

5.9 Signal LED Brightness

The brightness of LED signals is controlled by the amount of current through them. This in turn is controlled on the TurnoutBoss board by the "Signal LED Brightness" configuration control. Higher numbers are brighter, lower numbers are dimmer. The default is 6 which is about 75% of full current. Note there are four separate tabs for the four signals.

5.10 Servo Limits

When the turnout is commanded to move, the servo output moves between two positions. The default starting position for the servo output when set to normal is 85 of 255. The default starting position for the servo output when set to diverging is 170 of 255.

In this section, you can adjust those positions. Set them so that the points just align with the stock rails of the turnout. Don't set them too far over: if the servo has to exert ongoing force to push the points too far against the stock rails, the servo will heat up and eventually fail.

5.11 Optical Input

Some optical sensors are active when a train is present at the sensor. These typically work by reflecting light off the rolling stock.

Other optical sensors are active when a train is *not* present at the sensor. These typically work by having the rolling stock block light to the sensor.

This section lets you configure the TurnoutBoss for the type of optical sensor you have installed.

Note that if you have not connected an optical sensor, you should leave this setting at its default of "Train detected when input is low".

6 Making LCC Connections To The Rest Of The Layout

The TurnoutBoss's LCC connection lets you integrate it with the rest of the layout. This section discusses some ways to do that. See the appendix for more details on these.

6.1 Remote Turnout Control

In addition to local control via its inputs, you can also command the TurnoutBoss to operate the turnout it controls by producing events on the LCC network.

6.2 Connecting to a Control Panel

The TurnoutBoss produces and consumes events that are useful to a control panel

- It produces events that indicate the position of the turnout.
- It produces events that indicate occupancy of the monitored track segments.
- It produces events that nidicate what the signals are displaying. This includes both the exact lamp colors, and also the "At Stop", "Not at stop" indication often displayed on a CTC panel
- It consumes events that you can use to "hold" or "release" the signal logic for either leftward or rightward travel or both.

A Event Definitions

This appendix defines all the Events that are produced or consumed by a TurnoutBoss. To use one of these, copy the Event ID [5] and paste it into the appropriate event slot in the other node that you're configuring.

In the following (P) means that TurnoutBoss will produce these event IDs, and (C) means that the TurnoutBoss will consume these event IDs.

[5] Some configuration tools will only show numerical values for these Event IDs, others will allow you to give them names. Consult the documentation for the configuration tool you're using for more information.

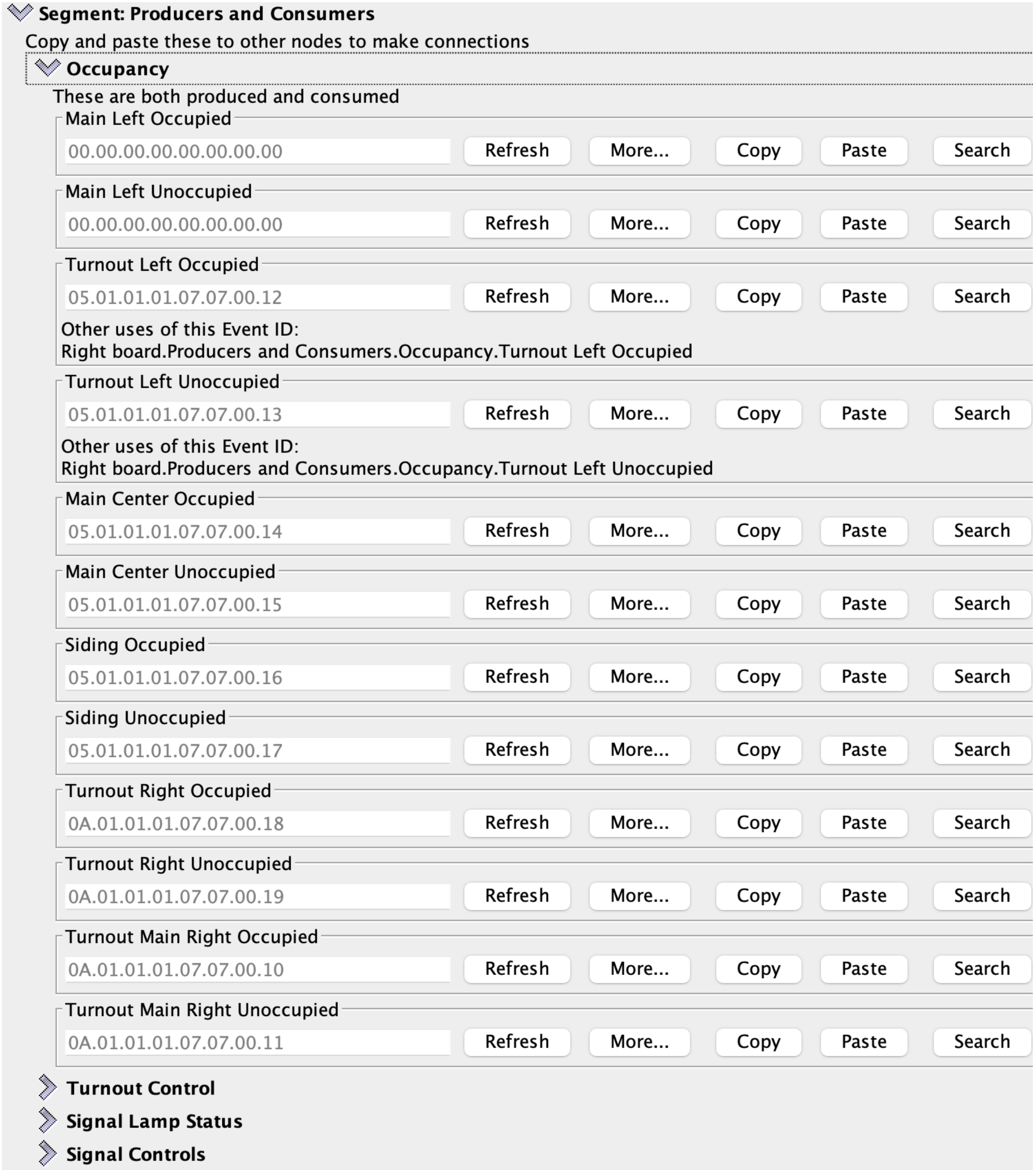


Figure 12: Producer and Consumer segment part 1

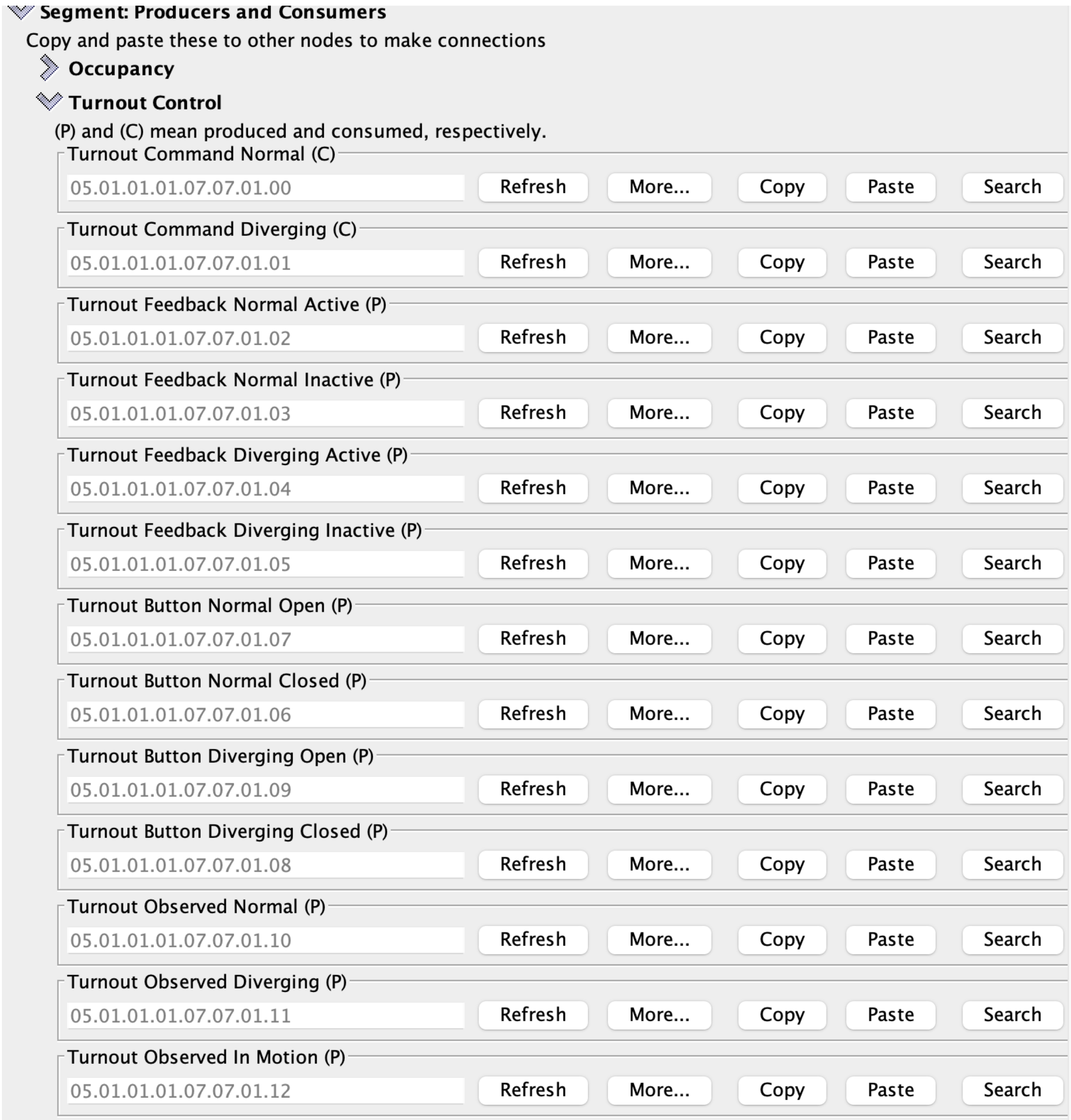


Figure 13: Producer and Consumer segment part 2

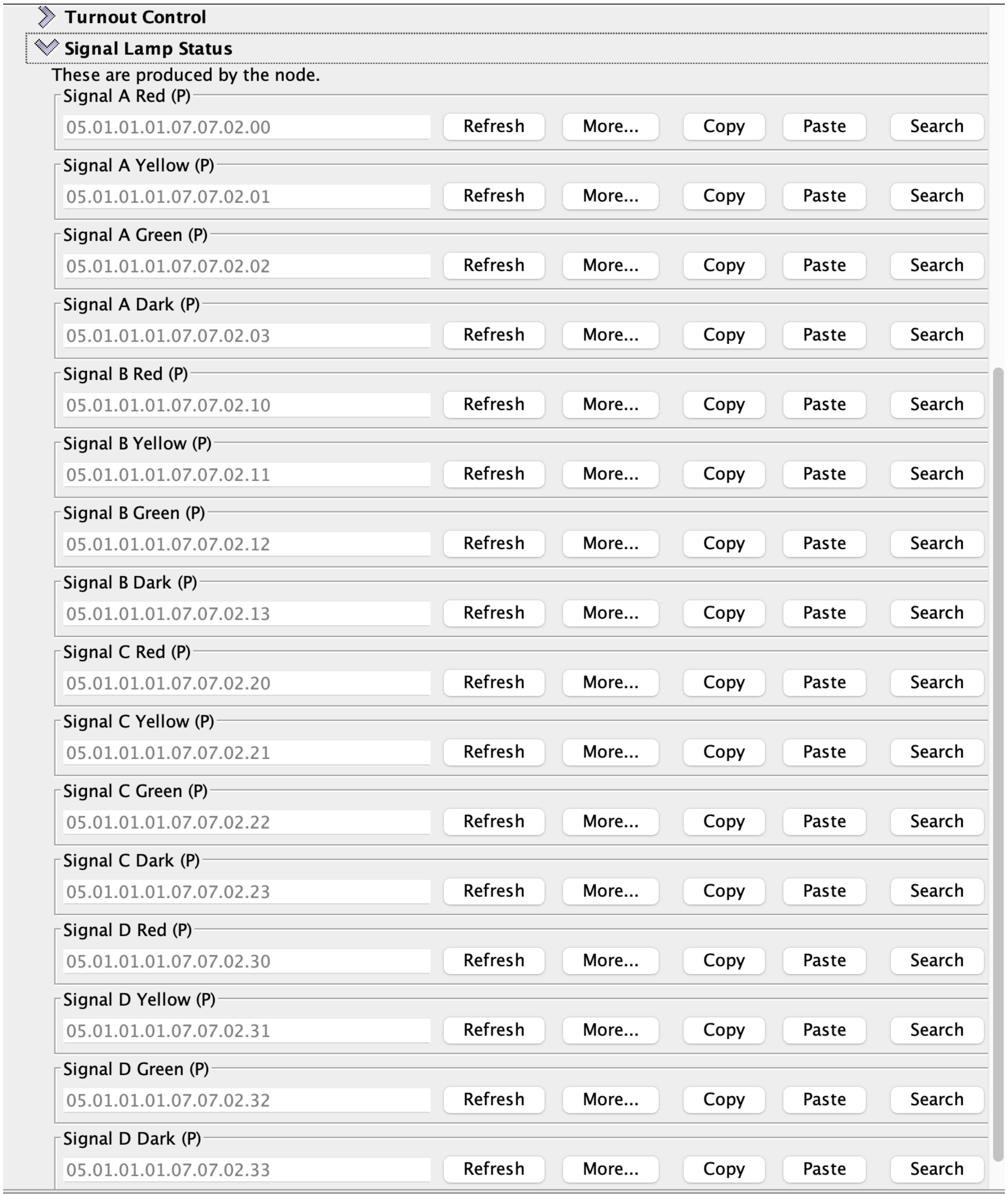


Figure 14: Producer and Consumer segment part 3

Figure 15: Producer and Consumer segment part 4

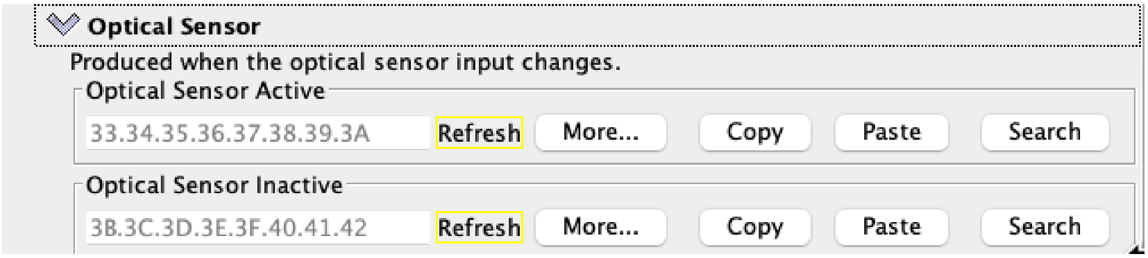


Figure 16: Producer and Consumer segment part 5

A.1 Occupancy

The TurnoutBoss will produce or consume these events when a track section becomes occupied or unoccupied. If the board is configured as a “Left” board it produces:

- Left Turnout
- Main Center
- Siding

and consumes

- Left Main

If the board is configured as “Right” it produces:

- Right Turnout
- Right Main

and consumes

- Main Center
- Siding

TurnoutBoss boards normally set up these producer to consumer connections automatically when you configure the Left-Right board pairs.

These events can also be used to indicate occupancy on a control panel or fascia indicators.

A.2 Turnout command normal and diverging (C)

When these events are produced somewhere else on the LCC network, the TurnoutBoss will consume them and change the position of the attached turnout. This lets you create one or more control panels that can control the turnout position.

A.3 Turnout Feedback Normal and Diverging, Active and Inactive (P)

There are two “Feedback” inputs to the TurnoutBoss board. The board can be configured to use one or both of them to feed back the position of the turnout, see above.

Alternately, the buttons can be used as general inputs. When the “Normal” input goes Active (closed) it will produce its Active event; when it goes Inactive (open), it will produce its Inactive event.

Similarly, the “Diverging” input will produce its Active and Inactive events when it goes Active and Inactive.

A.4 Turnout Button Normal and Diverging, Open and Closed (P)

There are two “button” inputs to the TurnoutBoss board. The board can be configured to use one or both of them to control the position of the turnout, see above.

In addition, the buttons can be used as general inputs. When the “Normal” button is pressed it will produce its Closed event; when it is released, it will produce its Open event.

Similarly, the “Diverging” button will produce its Closed and Open events when pressed and released.

A.5 Turnout Observed Position (P)

The TurnoutBoss produces events to confirm the motion and position of the attached turnout. You can configure the board to use zero, one or two switches to confirm the position of the board, see above.

When configured to zero or one switches, the board will produce the “Observed Normal” and “Observed Diverging” when the turnout position changes.

When configured to two switches, the board will also produce the “Observed In Motion” event when the first switch has opened and the second switch has not yet closed.

These can be used to drive the turnout indicators on a CTC control panel.

A.6 Signal Outputs (P)

The TurnoutBoss board will produce events when it changes its signal head outputs. For each of the A, B, C and D signals, there’s an event that corresponds to the signal going Green, Yellow, Red or Dark.

These can be used to put a repeater signal on the layout or on a control panel.

A.7 Signal State (P)

These are used by the two Turnout Boss boards around a single siding to communicate the status of their signals to each other.

The board will produce these events to indicate whether a signal is at stop (showing all Red) or clear (showing Green or Yellow). These can be used to drive the signal indicators on a CTC control panel.

A.8 Signal State (C)

These are used by the two Turnout Boss boards around a single siding to communicate the status of their signals to each other. There's generally no need for other boards on the LCC network to produce these events.

A.9 Vital Logic Control

These events can be used by a CTC panel to control the signal logic for the signals controlled by the TurnoutBoss board.

The last event consumed controls the operation:

- Both Clear - The signals in both directions will be cleared to green or yellow so long as the track ahead is clear. This is the initial state of the TurnoutBoss board so that the signals will allow traffic in both directions.
- Right Clear - The signals left-bound are held at stop. The signals right-bound will operate normally, showing stop if the track ahead is not clear.
- Left Clear - The signals right-bound are held at stop. The signals left-bound will operate normally, showing stop if the track ahead is not clear.
- Both Held - The signals in both directions will stay at stop. Trains will not be cleared to move in either direction.

These can be connected to a CTC Signal lever to control which trains are given permission to move.

A.10 Optical Sensor

These events are produced when the optical sensor on the turnout becomes active or inactive. This can also create occupancy events and changes to signals, etc.